Technical Report

VIRTUAL VAHANA 2026

Phase 1: Safety-first ADAS + Assisted Driving Systems

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1 Problem Statement & Objectives

The transition toward autonomous mobility necessitates robust, fail-safe intermediate systems. The primary objective of Phase 1 of the Virtual Vahana 2026 competition is to design, implement, and validate a comprehensive Advanced Driver Assistance System (ADAS) within a simulated environment.

As dictated by the official competition rules, fully autonomous navigation is strictly prohibited in this phase. The ego-vehicle must be driven manually using a keyboard or joystick, while the ADAS acts as an active safety net. The core engineering challenge lies in developing a system that accurately perceives its environment in real-time, evaluates collision risks, and correctly arbitrates control between the human driver and the safety systems without causing unsafe conflicts.

The system must successfully implement Lane Keeping Assist (LKA), Adaptive Cruise Control (ACC), Autonomous Emergency Braking (AEB), and Traffic Sign/Light Recognition, alongside our innovation feature: Blind Spot Detection (BSD).

2 System Architecture

Our proposed solution is a highly advanced, hybrid ADAS framework that marries Deep Reinforcement Learning for fluid decision-making with strict, hard-coded safety logic for feature arbitration. The architecture is deployed within the CARLA Simulator (v0.9.16).

2.1 Architecture Block Diagram

The following diagram illustrates our complete ADAS development architecture, detailing the flow from human inputs and simulator sensors through our A3C Decision Engine, into the Arbitration Layer, and finally to Control Actuation and the HMI.

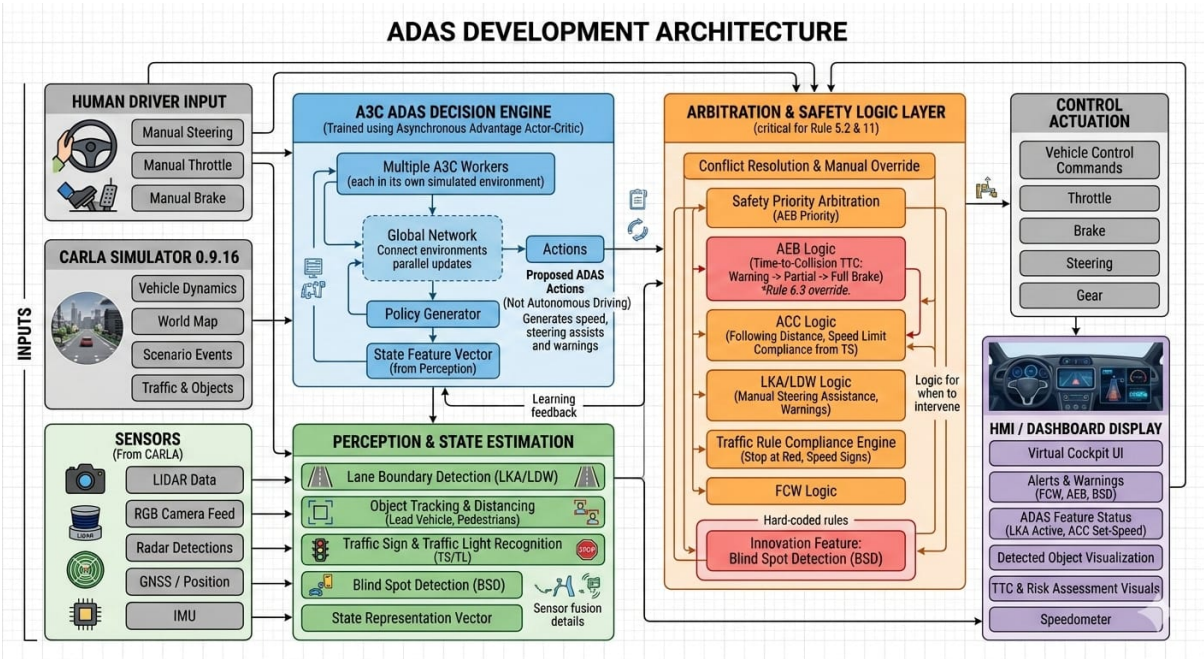

Figure 1: ADAS Development Architecture illustrating the A3C Decision Engine and Safety Arbitration Layer.

2.2 Subsystem Breakdown

The architecture is divided into five continuous pipelines:

1. **Inputs & Simulation:** Manual driving inputs (Steering, Throttle, Brake) are fed into the system alongside environment data from CARLA's robust sensor suite (LiDAR, RGB Cameras, Radar, GNSS, IMU).

2. **Perception & State Estimation:** Raw sensor data is fused into a State Representation Vector. This includes Lane Boundary Detection, Object Tracking, Traffic Sign/Light Recognition, and our GPS-based BSD.

3. **A3C ADAS Decision Engine:** A deep reinforcement learning model (Asynchronous Advantage Actor-Critic) generates proposed ADAS actions (speed limits, steering assists) based on the state vector.

4. **Arbitration & Safety Logic Layer:** The critical safety net. It evaluates the A3C's proposed actions against hard-coded safety rules (e.g., Rule 6.3 overrides) and manual inputs to resolve conflicts.

5. **Actuation & HMI:** The final arbitrated commands are sent to the vehicle's control actuators, while telemetry and warnings are displayed on the Virtual Cockpit UI.

3 Methodology: A3C Decision Engine

To generate smooth, human-like ADAS assistance, we implemented an Asynchronous Advantage Actor-Critic (A3C) reinforcement learning architecture. Traditional PID-only systems can struggle with edge cases in complex environments; A3C allows our system to learn optimal assistance policies across parallel simulated environments.

3.1 A3C Architecture

The A3C framework utilizes a global network and multiple worker agents. Each worker interacts with its own instance of the CARLA environment, calculating gradients and updating the global network asynchronously.

The policy gradient is updated using the advantage function $A(s_t, a_t)$:

$$\nabla_\theta J(\theta) \approx E\left[\nabla_\theta \log \pi_\theta(a_t|s_t) A(s_t, a_t)\right]$$

Where π_θ is the policy network (Actor) parameterized by θ. The advantage is calculated as the difference between the observed reward return and the value network's baseline estimate:

$$A(s_t, a_t) = R_t - V_{\theta_v}(s_t)$$

3.2 State Feature Vector

The Perception module feeds the A3C engine a condensed state vector S_t, containing:

- Ego-vehicle kinematics (v_x, v_y, yaw rate).
- Relative distancing to the lead vehicle ($\Delta x, \Delta v$).
- Lane offset error and heading angle.
- Active traffic sign constraints (e.g., speed limit scalar).

Crucially, the A3C engine does *not* generate fully autonomous driving commands, preserving compliance with Phase 1 rules. It generates *proposed* assists and warnings.

4 Arbitration & Safety Logic Layer

While the A3C engine handles fluid driving states, the **Arbitration & Safety Logic Layer** utilizes strict, deterministic rules to guarantee safety and compliance with Rule 5.2 (ADAS overriding the driver only when required for safety).

4.1 Time-to-Collision (TTC) & AEB Override

Autonomous Emergency Braking (AEB) operates independently of the A3C policy to guarantee instantaneous reaction times. Let x_{rel} be the relative distance and v_{rel} be the relative velocity. TTC is calculated continuously:

$$TTC = \frac{x_{rel}}{v_{rel}} \quad \forall\ v_{rel} > 0$$

The arbitration logic enforces a multi-stage approach:

- **Warning Stage ($TTC \leq 3.0s$):** Triggers FCW logic.
- **Partial Brake ($TTC \leq 2.0s$):** Overrides ACC, applies 40% braking capacity.
- **Full Brake ($TTC \leq 1.0s$):** As per Rule 6.3, AEB strictly overrides all manual throttle, manual braking, and A3C proposed actions to apply 100% braking force.

4.2 Traffic Rule Compliance Engine

This module hard-caps the throttle outputs allowed by the ACC logic. If the Perception layer detects a Red Light or a 40 km/h sign, the Compliance Engine forcefully scales down the maximum allowable throttle command sent to the Control Actuation block, regardless of the human driver's input.

5 Innovation Feature: Blind Spot Detection

Embedded within our Arbitration layer is our Innovation Feature: Blind Spot Detection (BSD). Using fused GNSS, IMU, and Radar data from CARLA, we track vehicles in adjacent lanes.

If a vehicle falls within the longitudinal range of $[-3.0m, +1.0m]$ relative to the ego-vehicle's rear axle, and within a lateral lane offset of $\pm 3.5m$, the target is flagged. If the human driver provides a manual steering input toward the occupied lane, the BSD logic instantly triggers the HMI alert and injects a counter-steering resistance into the Control Actuation layer.

6 Human-Machine Interface (Dashboard)

The HMI was developed as a Virtual Cockpit UI to provide real-time telemetry with low cognitive load on the driver.

Key UI features include:

- **ADAS Feature Status:** Displays the active state of LKA and the current ACC set-speed.
- **TTC & Risk Visuals:** A dynamic color-coded bar providing visual context for forward collision risks.
- **Alerts & Warnings:** High-contrast pop-ups for FCW, AEB activation, and BSD alerts, ensuring immediate driver awareness during critical interventions.

7 Testing Methodology & Results

The system was validated using CARLA Simulator 0.9.16. We tracked the system's performance across various traffic scenarios to evaluate the A3C engine's smoothness and the Arbitration Layer's safety enforcement.

7.1 Feature Arbitration Validation

Test Condition	**Arbitration Logic**	**Result**
ACC active, lead vehicle brakes hard	AEB detects TTC $< 1.0s$. Arbitration Layer overrides A3C; forces brake to 1.0.	Pass
Speed limit drops from 60 to 30 km/h	Camera detects 30 km/h sign. Compliance Engine caps ACC target speed.	Pass
Manual lane change, blind spot occupied	BSD detects target. HMI flashes. Counter-steering logic triggered.	Pass

Table 1: Validation of feature interaction and safety overrides.

8 Challenges & Future Scope

8.1 Integration Challenges

The primary challenge during Phase 1 was bridging the asynchronous nature of the A3C updates with the synchronous tick-rate of the CARLA simulation environment. Furthermore, tuning the Arbitration Layer to ensure AEB overrides felt distinct but not overly aggressive required significant empirical testing.

8.2 Transition to Phase 2

The integration of an A3C Decision Engine positions Team FireWarriors exceptionally well for Phase 2. Moving forward, we intend to expand the A3C's action space to include full trajectory generation, shifting the architecture from Assisted Driving to a fully Autonomous Driving System.